

Project: Build a Web-Based Calculator (Basic Mode – Windows 11 Equivalent)

Objective

You are required to develop a **web-based calculator** that replicates the **Basic Mode** functionality of the Windows 11 Calculator.

The calculator should support standard arithmetic operations (addition, subtraction, multiplication, division, percentage, square root, etc.), follow **modern web development practices**, and be implemented as either a **single-page web application (SPA)** or a **pure JavaScript app**.

Technical Requirements

You may choose **one** of the following development approaches:

- **Option 1:** Build using **Raw HTML, CSS, and JavaScript**
- **Option 2:** Build using a **modern SPA framework** (e.g., React, Vue, Svelte)

Your application must:

- Run fully in the **browser** (no backend).
- Have a **user-friendly interface** similar to Windows 11 Basic Mode.
- Be **responsive** and **visually organized**.
- Be **publicly hosted** on a platform such as:
 - GitHub Pages
 - Netlify
 - Vercel

- Firebase Hosting
- or any other public hosting provider

The hosted link must be included in your documentation.

Submission Requirements

Your documentation must include the following sections:

Functional Specifications

- Purpose and scope of your calculator.
- Full list of supported features (same as Windows 11 Basic Mode).
- Description of how user inputs, operators, and the display are handled.
- Include any assumptions (e.g., rounding, operator order).

Non-Functional Specifications

- Performance (quick response, smooth updates).
- Usability (clear layout, accessible for users).
- Cross-browser compatibility (Chrome, Edge, Firefox, Safari).
- Responsiveness (desktop & mobile support).
- Reliability and maintainability.

Acceptance Criteria

- Arithmetic operations return correct results.
- Operator precedence is correctly applied.

- CE, C, and Backspace behave as in Windows Calculator.
- Display updates accurately after each input.
- Design remains stable across browsers and devices.
- **Deployed version is publicly accessible and functional.**

Testing Plan

- Explain the testing method (manual or automated).
- Provide **test cases** for all main features, including:
 - Input
 - Expected Output
 - Actual Output
 - Result (Pass/Fail)
- Include example cases such as:
 - $2 + 3 = \rightarrow 5$
 - $\sqrt{9} = \rightarrow 3$
 - $10 \% \rightarrow$ behaves as 10% of current operand

Prompt Engineering (AI Assistance)

- Include any **AI prompts** used for code or documentation generation.
- Explain how AI helped in your development process (e.g., idea generation, debugging).
- Reflect on what you learned from using AI.
- All AI-generated content must be **reviewed and verified** by you.



Submission Format

Submit a **ZIP file** containing:

1. **Source code** of your web-based calculator.
2. **Documentation file** (`.pdf` or `.docx`) including all sections above.
3. The **public hosting link** clearly stated in the document.

👉 Follow the **submission instructions in the previous *Course Introduction* slide** for file naming and upload procedures.



AI Assistance Policy

You **may use AI tools** (e.g., ChatGPT, GitHub Copilot, Gemini) to help:

- Write documentation
- Generate or debug code
- Design UI or test cases

However:

- You **must review** and **take full responsibility** for all generated content.
 - Clearly mark which sections involved AI assistance.
 - Submissions with unreviewed AI content will lose marks.
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Suggested Tech Stack

Category

Recommended Tools

Frontend	HTML, CSS, JavaScript, React, Vue, or Svelte
Hosting	GitHub Pages, Netlify, Vercel, Firebase Hosting
Testing	Jest, Mocha, Cypress, or manual testing plan
Version Control	Git / GitHub (optional but encouraged)



Required Features (Basic Mode Only)

Feature	Description
Number Input	0–9, decimal point
Basic Operations	$+$, $-$, $\times$, $\div$
Percentage (%)	Compute percentages
Square Root ($\sqrt{}$)	Compute square root
Negate ($\pm$)	Toggle positive/negative
Clear Entry (CE)	Clear current input
Clear (C)	Clear all entries
Backspace ($\leftarrow$)	Delete last character
Display	Show current and previous values
Keyboard Support (<i>optional</i>)	Allow number and operator keys
Public Hosting	Deployed on a public hosting platform with a working demo



Grading Rubric

Criteria	Description	Weight
Functional Specifications	Completeness and accuracy of described calculator features	15%
Non-Functional Specifications	Clarity and coverage of usability, performance, and responsiveness	10%

Acceptance Criteria	Well-defined, testable, aligned with requirements	10%
Testing Plan	Clear test cases, expected results, organized structure	10%
Source Code Quality	Correctness, readability, structure, and best practices	25%
Public Hosting & Deployment	Calculator successfully deployed, accessible online, and functions as expected	10%
Prompt Engineering (AI Assistance)	Clear inclusion of prompts, responsible AI use, reflection quality	10%
Documentation & Presentation	Overall clarity, structure, and professionalism of the report	10%
Total: 100%		